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International A-Level

Physics (9630)
PH03
Report on the Examination

June 2026

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REPORT ON THE EXAMINATION: INTERNATIONAL A-LEVEL PHYSICS (9630) PH03 – JUNE 2026

In general, the paper was well answered with hardly any questions left blank. Students had enough time to consider the questions and to answer them in sufficient detail in written answer. Students also generally showed their working. The multiple choice section was very well answered, with many students scoring highly.

A large number of students did not use a ruler when drawing diagrams; on the occasions when this was needed, for example, for drawing vectors, students would probably have scored more highly if they had used rulers.

QUESTION A1.1

This proved to be a challenging question, with many students not scoring because they did not give enough detail. As in previous series tongs is not sufficient students should refer to *long tongs,* which is specific equipment that students should know about. Similarly, many students wrote about keeping a distance away without specifying the distance and so did not score.

QUESTION A1.2

This was generally well answered. Students did not gain the mark for using the phrase *possibility* or similar, instead of the mathematical word *probability*, which was required. Some did not get the mark because they were too vague in their answer and did not make it clear enough that it is the probability of a nucleus decaying. Many students answered in terms of the ratio of activity to the number of nuclei as an alternative, which was accepted. However, if an equation is used the symbols must be defined, which in this case a small number did not do.

QUESTION A1.3

This was well answered, with many students scoring well. However, many students did not take sufficient care to get an accurate value, either by taking multiple readings of half-life and averaging or by using a time difference greater than half of the maximum range possible from the graph.

QUESTION A1.4

This was well answered, although a number of students made the mistake of treating the number as being the number of carbon nuceli left, rather than the number of boron nuceli. This question discriminated well.

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QUESTION A2.1

The vast majority of students understood the correct arrow to draw but many did not gain credit because of badly drawn arrows: either not using a ruler or not connected to the ball. Most did label the arrow clearly, as asked.

QUESTION A2.2

The majority of students got a mark here, but many did not explicitly mention that weight was the force that required a vertical component to be present.

QUESTION A2.3

This question was very well answered, showing the high level of mathematical ability that these students have. The vast majority were very comfortable using trignometry to resolve the forces and solve the problem.

QUESTION A2.4

This question was very well answered, with the vast majority scoring the first two marks. The second two relating to the speed, students found harder. A large number did use an algebraic argument to justify this, with many choosing to give an equation relating speed to radius and angle. Students are free to use a mixture of algebraic arguments and words in their explanations.

QUESTION A3.1

This was well answered by very many students.

QUESTION A3.2

This was well answered. A small number of students used $F = m\omega^2 r$ from circular motion to calculate the answer. This was not given credit as this situation was not circular motion, although it happens to give the correct numerical value.

QUESTION A3.3

This question discriminated well and a good number achieving full marks. A number of students had the correct direction but could not explain clearly enough that the force needs to be in opposite direction to displacement for SHM.

QUESTION A3.4

This was a challenging question, which many students did well ; showing good working and achieving at least partial credit. A number of students drew a cosine or sine graph to help them visualise the cycle, which seemed to help them find an approach to solve the question.

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QUESTION A3.5

This was another challenging question, which many students also scored well on. A number of students recognised that for SHM to be true it requred the force to be proportional to the displacement, but many could not then apply this to the situations and how the cloth would affect this condition.

QUESTION A4.1

This was well answered by many students.

QUESTION A4.2

This was well answered by many students. Most students got full marks, including giving the negative sign. However, some did miss the negative sign. Some lost credit by giving an answer to 1 significant figure only. Calculation values should be given to at least 2 significant figures, unless the answer is constrained to be an integer or limited for another reason, as in A6.4

QUESTION A4.3

This was answered well by many students. A large number produced the algebraic equation for total energy, as shown in the guidance.

QUESTION A5.1

This was well answered by most students.

QUESTION A5.2

Many students correctly realised that they had to compare the distance moved by the spider with the radius or diameter of the Earth. Some did not give a comparson and so did not score the mark. A small number of students answered by explaining that the field lines were parallel. While this indicates that the field is uniform it does not explain why the electric field is uniform.

QUESTION A5.3

Many students correctly answered this by stating the electrostatic force is in the oppoiste direction to the electric field lines. Some answered by saying the electric field lines go from positive to negative, or are in the direction of a force on a positive charge, without doing on to relate this to why the spider should be negative.

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QUESTION A5.4

This was well answered. A number did not take the weight of the spider into account, and that the resultant force would be the difference between the electrostatic force and the weight. A large number of students used 9.8 for *g* rather than 9.81. This was condoned in this case, but it is expected that the students use the data book value of 9.81 and they should be encouraged to use data book values rather than recalling lower precision values.

QUESTION A6.1

Many students scored at least 1 mark for stating that the dielectric would reduce the electric field strength. However, developed explanations for why this occurs was rarer. Not many students explained this in terms of the alignment of polar molecules with the electric field.

QUESTION A6.2

This was well answered by many students. A small number gave a unit with their answer which prevented the award of 2 marks.

QUESTION A6.3

This was well answered by very many students. Students argued well using the capacitance equation to support their reasoning.

QUESTION A6.4

This was a challenging question and it discriminated well. Many calculated the time-base using the time period from the graph. A smaller number managed to calculate the voltage-gain. The majority of students recognised that they could calculate the peak voltage, but were then unsure how to use this to determine the voltage-gain. Some students solved the question but then gave an answer to more significant figures. In this case, the answer should match the settings on the dial shown and therefore e.g. 1.0 ms / div is not appropriate.

QUESTION A7.1

This was well answered by many students.

QUESTION A7.2

Many students could answer this question correctly, with most referencing Fleming's Left Hand Rule. There was no need to draw a labelled diagram of the rule, but it did need to be named.

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QUESTION A7.3

This was well answered, but to get the second mark, students needed to give an algebraic explanation. Many did this well by deriving or stating the equation for the radius and linking this to the mass of the charged particle.

QUESTION A7.4

This question discriminated well. Many students did get full marks, because they did not correctly use the SI prefixes on the graph axes, or they used the diameter instead of the radius.

QUESTION A7.5

This question was answered well, but a number of students made miscalculations or rounding errors.

QUESTION A7.6

Many students seemed to understand the physics, but a number did not score because of underdeveloped answers e.g. using ‘equal and opposite forces’ or ‘balanced forces’ without explicitly saying that the electric force and the magnetic force would be in opposite directions. Often, the minimalist answer of ‘equal and opposite’ will not be accepted as sufficient detail.

SECTION B

Most of the multiple-choice questions were answered well, many by over 75% of the students.

Questions that were found to be harder were 15, 19 and 21. In these cases, most students got the questions correct, but in smaller amounts than above.

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